IDSC 2026 Pitch Deck

Helping clinicians miss fewer Brugada cases

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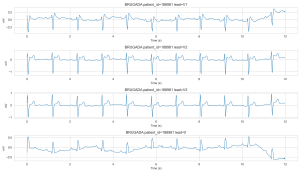


A safety-aware ECG benchmark for earlier review of high-risk patients.

One missed Brugada ECG can be fatal

Why judges should care

- Brugada syndrome can lead to sudden cardiac death.
- In screening, missing a true case matters more than one extra review.
- So the key question is: "Who are we still missing?"

Our story

We use ordinary 12-lead ECGs to build a screening aid designed to miss fewer dangerous patients.

356

patients in the main ECG cohort

69

Brugada cases in the full benchmark

11

models on the same ECG-only split

We benchmarked for safety, not just score

What we did

- Read raw 12-lead ECGs.
- Compare 11 models on one shared patient split.
- Keep the main benchmark ECG-only.

Clinical design choice

Metadata stays in a separate ablation, so the main result is driven by the ECG itself.

How we judged models

- Strong overall performance still matters.
- But missing risky patients matters more.
- Consistency and practicality break ties.

Key message

We do not let a high score hide dangerous false negatives.

We did not bet on one lucky algorithm

Three model families

- **Feature models**: XGBoost, Random Forest, SVM-RBF, and AdaBoost.
- **Deep sequence models**: CNN, ResNet-1D, CNN+BiGRU, and Transformer Encoder.
- **Representation learning**: transfer learning, VICReg, and Echo State Network.

Why this matters

- Each family reads the ECG differently.
- All 11 models use the same split, so the comparison is fair.
- If one model still wins, the recommendation is easier to trust.

Takeaway: we tested breadth, then chose the model that balances performance with screening safety.

One model won on safety and performance

Recommended model

Transfer learning model

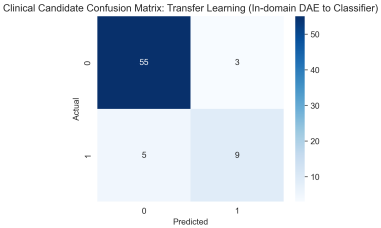
- **Headline metric:** PR-AUC 0.676
- **Clinical takeaway:** only 5 false negatives

The story in one sentence

The best leaderboard model and safest screening model can differ. Here, they converge on the same transfer model.

Why this result matters

This is what we want in healthcare: a model that performs strongly overall while still reducing the number of missed risky patients.



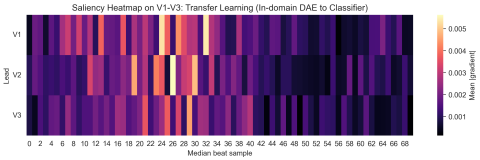
It fits the way clinicians already work

Why it is practical

- It uses standard 12-lead ECGs already available in care settings.
- It works as a second reader for earlier specialist attention.
- It still highlights plausible precordial regions.

Why it is clinically honest

- We explicitly surface false negatives.
- We separate aggregate performance from safety decisions.
- We frame it as a second reader, not autonomous diagnosis.



Simple value: help clinicians notice risky ECGs earlier.

From ordinary ECGs to earlier action

What we achieved

We built a safety-aware ECG benchmarking pipeline that turns raw 12-lead signals into a practical Brugada screening aid.

Near-term value

- Triage support
- Second-reader assistance
- Better prioritization for specialist review

Next step

- External validation
- Prospective threshold calibration
- Real clinical workflow integration

Miss fewer risky patients. Review them earlier.

That is the kind of impact AI should bring to healthcare.

Project repository: `https://github.com/avezoor/IDSC2026`